NATIONAL DEFENCE ACADEMY (NDA)

PGD PROGRAMME – ELECTRONICS / ELECTRICAL & ELECTRONICS ENGINEERING

FULL DETAILED LECTURE NOTE

INTRODUCTION TO SEMICONDUCTOR MATERIALS

Elemental vs. Compound Semiconductors: Si, Ge and GaAs

Crystal Structures • Miller Indices • Bonding Mechanisms

Prepared by

LECTURE NOTE

Purpose: Comprehensive classroom, examination and revision material for PGD students studying semiconductor materials and solid-state electronics.

Text book

* fundamental of Solid State Electronic ics II-C-Sah world 1991
* Solid state Electronic Circuits by Edward w maby. University of Southern California.
* Solid State Electronic Devices Ben G. Streetman. and Sanjay Kumar Banerjee.

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Solid State Electronics – ELE717

According to the **National Universities Commission (NUC)** guidelines for Engineering in Nigeria—including the traditional **BMAS** (Benchmark Minimum Academic Standards) and the updated **CCMAS** (Core Curriculum and Minimum Academic Standards)—**Solid State Electronics** is a 3-credit unit compulsory course.

The curriculum explicitly bridges fundamental physics with active semiconductor device technology.

Core Course Content Breakdown

- Physics & Material Properties of Semiconductors: Elemental and compound semiconductor materials; energy bands; carrier concentrations and statistics; high-field effects and velocity saturation.
- Carrier Transport & Injection Phenomena: Diffusion and drift currents; carrier lifetime; excess carriers; carrier injection mechanisms and semiconductor surface phenomena.
- Material Growth & Impurity Control: Bulk and epitaxial material growth techniques; purification of silicon; crystal growth; doping mechanisms and impurity control.
- Junctions & Interfaces: Metal-semiconductor interface properties; Schottky barriers versus Ohmic contacts; stability and methods of characterisation.
- Device Technology & Application Principles: Physics of operational solid-state electronic devices including standard p-n junction diodes, bipolar junction transistors (BJTs), and Field-Effect Transistors (FETs/MOSFETs).

Expected Learning Outcomes

By completing this curriculum, engineering students must be able to:

- Differentiate crystal structures and bonding mechanisms unique to electronic materials.
- Formulate mathematical models for carrier transport, diffusion, and injection across junctions.
- Analyze the operational limits of transistors influenced by high-field effects or interface instabilities.
- Evaluate modern device fabrication processes like epitaxy and impurity control.

***Course* Description**

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1. INTRODUCTION TO SEMICONDUCTOR MATERIALS

Definition: A semiconductor is a material whose electrical conductivity is intermediate between that of a good conductor and an insulator, and whose conductivity can be controlled by temperature, illumination, electric field, magnetic field and impurity concentration.

The importance of semiconductors arises from the fact that their electrical properties can be deliberately engineered. This controllability makes them suitable for switching, amplification, rectification, sensing, energy conversion, signal processing and computation.

A semiconductor is not simply defined by having a particular conductivity value. Its most useful characteristic is the existence of a controllable energy gap between occupied and unoccupied electronic states.

The principal semiconductor materials studied at introductory postgraduate level are silicon (Si), germanium (Ge) and gallium arsenide (GaAs). Other technologically important materials include gallium nitride (GaN), indium phosphide (InP), silicon carbide (SiC), cadmium telluride (CdTe) and indium gallium arsenide (InGaAs).

Scope of this note: The present lecture concentrates on material classification, atomic structure, crystal structures, bonding and Miller indices, while also introducing the material properties needed for later study of semiconductor devices.

2. HISTORICAL DEVELOPMENT AND IMPORTANCE OF SEMICONDUCTORS:

Early electrical experiments showed that certain materials did not behave like either perfect conductors or insulators. The development of solid-state physics later provided the theoretical basis for understanding these materials.

The invention of the transistor in 1947 established semiconductor materials as the foundation of modern electronics. Silicon subsequently became dominant because of its abundance, thermal stability, mature fabrication technology and excellent silicon-dioxide interface.

Compound semiconductors became increasingly important where silicon could not provide optimum performance, particularly in high-frequency electronics and optoelectronics. GaAs became a major material because of its high electron mobility and direct band gap.

Today, semiconductor materials are central to computers, smartphones, communication systems, radar, satellite systems, electronic warfare equipment, navigation systems, sensors, renewable-energy systems and control systems.

3. ATOMIC STRUCTURE AND VALENCE ELECTRONS

An atom consists of a positively charged nucleus containing protons and neutrons, surrounded by electrons occupying quantised energy states.

Electrons in the outermost occupied shell are called **valence electrons**. They are particularly important because they participate in chemical bonding and determine many electrical properties of solids.

Silicon and germanium are Group 14 elements. Each has four valence electrons. In a crystal, each atom forms four strong directional bonds with neighbouring atoms.

Gallium belongs to Group 13 and arsenic to Group 15. In GaAs, the combination of three-valence-electron gallium and five-valence-electron arsenic gives an average of four valence electrons per atom, enabling a tetrahedrally coordinated semiconductor crystal.

4. CLASSIFICATION OF MATERIALS

Materials are commonly classified electrically as conductors, semiconductors and insulators. The distinction is explained more fundamentally using electronic energy bands.

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Conductors: Have many readily available mobile charge carriers. Their valence and conduction states overlap or a band is partially filled.

Semiconductors: Have a finite band gap small enough that thermal or other energy can create mobile carriers in useful concentrations.

Insulators: Generally possess a large band gap, making the thermal generation of mobile carriers very small at ordinary temperatures.

Class	Typical behaviour	Band-gap concept	Examples
Conductor	High conductivity	Bands overlap / partially filled	Cu, Al, Ag
Semiconductor	Moderate and controllable	Small-to-moderate band gap	Si, Ge, GaAs
Insulator	Very low conductivity	Large band gap	Glass, diamond, ceramics

5. ENERGY BANDS AND BAND GAP

In an isolated atom, electrons occupy discrete energy levels. When a very large number of atoms are brought together to form a crystal, interactions between atoms cause these levels to split into closely spaced allowed energy states called bands.

Valence band: The highest energy band that is substantially occupied by electrons at absolute zero.

Conduction band: The next higher allowed band in which electrons can move through the crystal and contribute significantly to electrical conduction.

Forbidden energy gap (Eg): The energy interval between the top of the valence band and the bottom of the conduction band in an ideal semiconductor.

For a semiconductor, the band gap is a fundamental material property. Silicon has an indirect band gap of about 1.12 eV at room temperature, germanium about 0.66 eV, and GaAs about 1.42 eV.

Useful conversion: $1\ \text{eV} = 1.602 \times 10^{-19}\ \text{J}$.

6. INTRINSIC AND EXTRINSIC SEMICONDUCTORS

Intrinsic semiconductor: A very pure semiconductor in which electron and hole concentrations are generated primarily by the semiconductor itself. In thermal equilibrium, $n = p = n_i$.

Extrinsic semiconductor: A semiconductor whose carrier concentration has been deliberately modified by adding impurities.

n-type: Donor impurities provide additional electrons. Electrons are majority carriers and holes are minority carriers.

p-type: Acceptor impurities create holes. Holes are majority carriers and electrons are minority carriers.

Doping does not normally destroy the basic crystal structure when performed at controlled concentrations. Instead, impurity atoms occupy substitutional or interstitial positions and introduce new electronic states.

Mass-action relationship at thermal equilibrium: $n\,p = n_i^2$.

7. ELEMENTAL SEMICONDUCTORS

An elemental semiconductor is composed principally of one chemical element. The most historically important elemental semiconductors are silicon and germanium.

Both Si and Ge belong to Group 14 and form tetrahedrally coordinated diamond-cubic crystals under ordinary conditions.

Elemental semiconductors are useful because their bonding and crystal structures are relatively well understood. Silicon in particular supports highly developed oxidation, lithography, diffusion, implantation and integrated-circuit

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processes.

8. SILICON (Si)

Silicon has atomic number 14 and four valence electrons. It crystallises in the diamond-cubic structure under ordinary semiconductor processing conditions.

Band gap: Approximately 1.12 eV at room temperature and indirect in nature.

Major advantages: High natural abundance, relatively low cost, good thermal stability, strong Si–Si bonding and the ability to form a high-quality native oxide (SiO_2).

Applications: Digital integrated circuits, microprocessors, memories, power electronics, sensors, solar cells, MOS devices and control electronics.

Important limitation: Silicon is an indirect-band-gap semiconductor, so it is less efficient than direct-band-gap materials for light emission.

9. GERMANIUM (Ge)

Germanium has atomic number 32 and four valence electrons. It also forms a diamond-cubic crystal structure.

Band gap: Approximately 0.66 eV at room temperature and indirect.

Advantages: High carrier mobility, which can be useful in high-speed applications.

Limitations: Higher intrinsic carrier concentration and greater temperature sensitivity than silicon; its native oxide is not as technologically useful as SiO_2.

Applications: Specialised high-speed devices, infrared optics and selected heterostructure technologies.

10. COMPOUND SEMICONDUCTORS

Compound semiconductors contain two or more elements and may be grouped according to periodic-table families. Examples include III–V, II–VI and IV–IV compounds.

III–V semiconductors: Formed from Group 13 and Group 15 elements, such as GaAs and InP.

II–VI semiconductors: Formed from Group 12 and Group 16 elements, such as CdTe and ZnSe.

Compound semiconductors offer considerable flexibility in band-gap engineering, lattice matching and optical properties. This makes them especially important in high-speed electronics and optoelectronics.

11. GALLIUM ARSENIDE (GaAs)

GaAs is a III–V compound semiconductor composed of gallium and arsenic. Its atoms are tetrahedrally coordinated and its common cubic form is zinc blende.

Band gap: Approximately 1.42 eV at room temperature and direct.

Electrical advantages: High electron mobility and favourable high-frequency transport characteristics.

Optical advantage: The direct band gap makes radiative recombination efficient, supporting LEDs and semiconductor lasers.

Applications: Microwave circuits, RF amplifiers, high-frequency transistors, satellite communication, radar, optoelectronic transmitters, LEDs, lasers and photovoltaic devices.

Limitations: More expensive and mechanically more fragile than silicon, with a less convenient native oxide and more complex manufacturing ecosystem.

12. DETAILED COMPARISON OF Si, Ge AND GaAs

The following comparison is useful for examination and engineering material-selection questions.

Property	Si	Ge	GaAs
Type	Elemental	Elemental	III–V compound
Crystal	Diamond cubic	Diamond cubic	Zinc blende
Band gap at ~300 K	≈1.12 eV	≈0.66 eV	≈1.42 eV
Band gap type	Indirect	Indirect	Direct
Valence electrons	4	4	Average 4/atom
Electron mobility	Moderate	High	High
Native oxide	Excellent SiO_2	Less useful	Not comparable to SiO_2
Major use	ICs, power, sensors	Specialised high-speed/IR	RF, microwave, optoelectronics
Relative cost	Low/moderate	Higher/specialised	Generally higher

13. CRYSTAL LATTICE AND UNIT CELL

A crystal lattice is an ideal mathematical array of points repeated periodically in space. A basis or motif is attached to each lattice point to generate the actual crystal structure.

The **unit cell** is the smallest repeating volume from which the entire crystal can be generated by translation.

A general unit cell is described by three lengths a, b and c and three angles alpha, beta and gamma.

Lattice constant: In a cubic crystal, the edge length a is commonly called the lattice constant.

Coordination number: Number of nearest neighbours surrounding each atom.

Atomic packing factor: APF = volume occupied by atoms / total unit-cell volume.

14. COMMON CRYSTAL STRUCTURES

Important idealised crystal structures include simple cubic (SC), body-centred cubic (BCC), face-centred cubic (FCC), diamond cubic and zinc blende.

Diamond cubic and zinc blende are particularly important for semiconductor materials because they provide tetrahedral coordination.

The conventional unit cell and the primitive unit cell are not identical concepts. A conventional cell is selected for convenience and symmetry; a primitive cell contains exactly one lattice point.

Structure	Atoms per conventional cell	Coordination number	APF (approx.)
Simple cubic	1	6	0.52
BCC	2	8	0.68
FCC	4	12	0.74
Diamond cubic	8	4	0.34

15. DIAMOND-CUBIC STRUCTURE

Silicon and germanium possess the diamond-cubic structure. Geometrically, the structure can be viewed as two identical FCC lattices displaced relative to one another by one-quarter of the body diagonal.

Each atom has four nearest neighbours arranged approximately at the corners of a tetrahedron. The bonding is strongly directional and covalent.

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The conventional cubic unit cell contains 8 atoms. The nearest-neighbour distance is related to the lattice constant by $r_{nn} = (\sqrt{3}/4)a$.

The diamond-cubic structure is not the same as ordinary FCC even though its underlying Bravais lattice is FCC. The additional basis creates the diamond structure.

For Si, using a = 5.43 Å: nearest-neighbour distance = (sqrt(3)/4)(5.43) = 2.35 Å.

16. ZINC-BLENDE STRUCTURE

GaAs commonly crystallises in the zinc-blende structure. It is structurally related to diamond cubic, but the two interpenetrating FCC sublattices contain different species.

One sublattice is occupied by gallium atoms and the other by arsenic atoms. Each Ga atom has four nearest-neighbour As atoms, and each As atom has four nearest-neighbour Ga atoms.

Because the two atoms are different, the crystal has polarity and mixed covalent–ionic character. This difference strongly affects electronic and optical properties.

17. WURTZITE STRUCTURE

Wurtzite is a hexagonal crystal structure also based on tetrahedral coordination. It is common in several compound semiconductors, including GaN.

Compared with zinc blende, wurtzite has a hexagonal stacking sequence rather than cubic stacking. The structural difference can affect polarization, strain, optical transitions and device behaviour.

Understanding both structures is important when studying modern wide-band-gap and optoelectronic semiconductor devices.

18. BONDING MECHANISMS IN SEMICONDUCTORS

Covalent bonding: Atoms share valence electrons. It is the dominant bonding picture for Si and Ge.

Ionic bonding: Electrons are transferred substantially between atoms, producing oppositely charged ions. Pure ionic bonding is not the normal description of Si, Ge or GaAs, but ionic character may be significant.

Metallic bonding: Valence electrons are delocalised over a lattice of positive ion cores. This explains the high conductivity of metals and contrasts with semiconductor bonding.

Van der Waals bonding: Weak interactions between neutral atoms or molecules. These are important in layered or molecular materials but are not the primary bond in conventional Si or GaAs crystals.

Mixed bonding: Many real semiconductor compounds have bonding that lies between ideal covalent and ionic limits.

19. COVALENT BONDING IN Si AND Ge

Silicon has four valence electrons and forms four covalent bonds with four neighbouring atoms. Each bond is formed by sharing electrons between the participating atoms.

At very low temperature, the ideal crystal can be pictured as having its bonding electrons localised in valence states. Thermal energy can break some bonds and create mobile electrons and holes.

The directional nature of covalent bonds explains the tetrahedral coordination and contributes to the mechanical and electronic characteristics of diamond-cubic semiconductors.

20. IONIC AND MIXED IONIC-COVALENT BONDING

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The electronegativity difference between different atoms causes charge redistribution. In GaAs, arsenic is more electronegative than gallium, so the electron distribution is not perfectly symmetric.

This partial charge separation contributes to the polarity of the GaAs bond. Nevertheless, electrons remain significantly shared, so GaAs is not treated as a purely ionic solid.

Bond character influences lattice energy, phonon behaviour, dielectric response, band structure, defect formation and optical transitions.

21. MILLER INDICES: CRYSTAL PLANES

Miller indices are integer indices used to identify the orientation of crystal planes. A plane is written as (h k l).

Procedure: (1) Find the intercepts of the plane with x, y and z axes in units of a, b and c. (2) Take reciprocals. (3) Clear fractions. (4) Reduce to the smallest integers. (5) Write the result in parentheses.

If the plane is parallel to an axis, its intercept on that axis is infinity and the reciprocal is zero.

If an intercept is negative, the corresponding index is written with a bar. For plain-text calculations this may be represented as a negative integer.

Example: intercepts (1, 2, 3) → reciprocals (1, 1/2, 1/3) → multiply by 6 → (6, 3, 2).

22. MILLER INDICES: CRYSTAL DIRECTIONS

A crystallographic direction is written in square brackets [u v w].

To determine a direction, subtract the coordinates of the starting lattice point from those of the final lattice point. Clear fractions and reduce the resulting components to the smallest integers.

Directions are vectors, while Miller indices of planes describe orientations. In cubic crystals, a direction [h k l] is normal to the plane (h k l), but this convenient relationship is not generally valid for non-cubic crystals.

[100] is along the x-axis; [010] along y; [001] along z; [110] lies in the x-y plane; [111] is the body-diagonal direction.

23. FAMILIES OF PLANES AND DIRECTIONS

A family of equivalent planes is written {h k l}. A family of equivalent directions is written <u v w>.

For example, in a cubic crystal the planes (100), (010) and (001), together with their symmetry equivalents, form the {100} family.

The distinction is important: (100) refers to one specific orientation, while {100} refers to all symmetry-equivalent orientations.

24. INTERPLANAR SPACING

Interplanar spacing d_{hkl} is the perpendicular distance between adjacent parallel planes belonging to a particular set of Miller indices.

For a cubic crystal with lattice constant a:

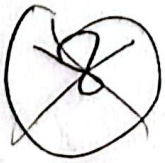
$$d_{hkl} = a / \text{sqrt}(h^2 + k^2 + l^2)$$

This equation shows that higher Miller indices generally correspond to smaller spacing for the same lattice constant.

25. WORKED MILLER-INDEX EXAMPLES

Example 1: Intercepts 2a, 3b, 1c. Reciprocals are 1/2, 1/3, 1. Multiply by 6: (3 2 6). Answer: (326).

Example 2: Intercepts a, b, infinity. Reciprocals are 1, 1, 0. Answer: (110).



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Example 3: Intercepts infinity, b, c. Reciprocals are 0, 1, 1. Answer: (011).

Example 4: Intercepts 2a, infinity, 2c. Reciprocals are 1/2, 0, 1/2. Multiply by 2: (101).

Example 5: A direction from (0,0,0) to (1,2,1) is [121].

Example 6: A direction from (1,1,1) to (0,2,3) has coordinate changes (-1,1,2), giving [-1 1 2].

26. CRYSTAL ORIENTATION AND SEMICONDUCTOR PROCESSING

Crystal orientation is fundamental in semiconductor fabrication. A wafer surface is identified by its crystallographic plane, such as (100) or (111).

Different crystal planes can have different surface atom densities, oxidation behaviour, etching rates, surface states and defect structures.

Anisotropic etching is particularly sensitive to crystal orientation because chemical reaction rates can differ from one plane to another.

Epitaxial growth requires careful control of lattice orientation and lattice matching between the substrate and deposited layer.

Cleavage planes can also be related to crystal structure and bonding. The ease of cleavage depends on the direction and strength of bonds in the lattice.

27. DEFECTS IN SEMICONDUCTOR CRYSTALS

A real crystal is never perfectly periodic. Defects strongly influence semiconductor behaviour.

Point defects: Vacancies, interstitials and substitutional impurities.

Line defects: Dislocations, which are extended imperfections in the crystal lattice.

Planar defects: Grain boundaries, stacking faults and interfaces.

Volume defects: Voids, precipitates and inclusions.

Defects can act as scattering centres, recombination centres, traps or diffusion paths. Some defects are harmful, while carefully controlled impurities are deliberately introduced to obtain desired electrical properties.

28. DOPING AND IMPURITY ATOMS

Doping is the deliberate addition of controlled quantities of impurity atoms to modify semiconductor electrical properties.

Donor doping: A Group V atom such as phosphorus in silicon has five valence electrons. Four can participate in covalent bonds, leaving one electron relatively weakly bound and available for conduction.

Acceptor doping: A Group III atom such as boron has three valence electrons. Replacing a silicon atom leaves an incomplete bond, which can be described as a hole in the valence system.

Doping concentration must be carefully controlled. Excessive impurity concentration can cause unwanted scattering, compensation and changes in semiconductor band structure.

29. MATERIAL SELECTION FOR ELECTRONIC APPLICATIONS

Material selection is based on electrical, optical, thermal, mechanical, chemical and manufacturing requirements.

Choose Si when: large-scale integration, low cost, mature processing, good oxide quality and power-device capability are important.

Choose GaAs when: high-frequency performance, high electron mobility or efficient light emission is important.

Choose Ge or Ge-containing structures when: high mobility or infrared-related properties provide a particular advantage.

In advanced engineering, material selection is a trade-off involving band gap, mobility, thermal conductivity, dielectric properties, breakdown field, defect density, cost and manufacturability.

30. SUMMARY AND KEY POINTS

A semiconductor is a material with controllable electrical conductivity and a finite energy band gap.

Si and Ge are elemental Group 14 semiconductors with diamond-cubic structures.

GaAs is a III–V compound semiconductor with a zinc-blende structure in its common cubic form.

Crystal structure determines how atoms are arranged in three dimensions and strongly affects semiconductor properties.

Bonding in Si and Ge is predominantly covalent; GaAs has substantial covalent character with partial ionic character.

Miller indices provide a systematic notation for crystal planes and directions.

For cubic crystals, $d(hkl) = a/\sqrt{h^2+k^2+l^2}$.

Crystal orientation is essential in wafer processing, etching, epitaxy and device fabrication.

Defects and impurities can strongly influence mobility, lifetime, recombination, conductivity and reliability.

31. WORKED NUMERICAL PROBLEMS

Problem 1 – Interplanar spacing: A cubic semiconductor has lattice constant a = 5.43 Å. Calculate d(111).

Solution: $d(111) = 5.43/\sqrt{1+1+1} = 5.43/\sqrt{3} = 3.135$ Å.

Problem 2 – d(220): Let a = 5.65 Å. Calculate d(220).

Solution: $d(220) = 5.65/\sqrt{4+4+0} = 5.65/\sqrt{8} = 1.998$ Å.

Problem 3 – Miller plane: A plane has intercepts 1a, 2b and 4c. Determine its Miller indices.

Solution: Reciprocals = (1, 1/2, 1/4). Multiply by 4 → (4,2,1). Therefore the plane is (421).

Problem 4 – Plane parallel to y: Intercepts are 2a, infinity, 1c. Reciprocals = (1/2,0,1). Multiply by 2 → (101).

Problem 5 – Direction: A direction starts at (1,0,1) and ends at (3,2,0). Difference = (2,2,-1), so the direction is [22-1].

Problem 6 – Nearest-neighbour distance: For diamond cubic, $r = (\sqrt{3}/4)a$. If a = 5.43 Å, r = 2.35 Å.

32. EXAMINATION AND REVISION QUESTIONS

Short-answer questions:

- Define a semiconductor.
- State three factors that can change semiconductor conductivity.
- Differentiate between intrinsic and extrinsic semiconductors.
- List two elemental and three compound semiconductors.
- Why is silicon widely used in integrated circuits?
- Why is GaAs important in optoelectronics?
- Define a crystal lattice and unit cell.

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- State the coordination number of diamond cubic.
- Explain the difference between diamond cubic and zinc blende.
- Define covalent, ionic and metallic bonding.
- What are Miller indices?
- Explain the meaning of (111), [111], {111} and <111>.
- Why does an infinite intercept produce a zero Miller index?
- Derive or state the cubic interplanar-spacing equation.
- Explain the importance of crystal orientation in wafer processing.
- What are point defects?
- Differentiate donor and acceptor impurities.
- Compare the band-gap types of Si, Ge and GaAs.

Long-answer / essay questions

- Discuss semiconductor materials and explain why their electrical properties can be controlled.
- Compare Si, Ge and GaAs with respect to structure, bonding, band gap, advantages, limitations and applications.
- Explain the diamond-cubic structure of silicon with reference to coordination and bonding.
- Explain the zinc-blende structure of GaAs and discuss its significance in optoelectronics.
- Give a detailed procedure for determining Miller indices and solve three examples.
- Discuss the role of crystal orientation in semiconductor manufacturing.
- Explain how bonding, defects and doping influence semiconductor properties.

33. SUGGESTED FURTHER READING

S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*.

Ben G. Streetman and Sanjay Banerjee, *Solid State Electronic Devices*.

Donald A. Neamen, *Semiconductor Physics and Devices*.

Robert F. Pierret, *Semiconductor Device Fundamentals*.

Neil W. Ashcroft and N. David Mermin, *Solid State Physics*.

Charles Kittel, *Introduction to Solid State Physics*.

END OF FULL DETAILED LECTURE NOTE

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